

D

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We DO NOT USE connector 3 or 4
Connector 3 has 16 GTY and 8 GTH
Connector 4 has 8 GTH
They're awkward connectors (AcceRate, essentially BGA)
and we don't need them (theoretically)
Worst case in rev 1 we drop down to GbE or something and add them later

SOM_B2B_1



B2B 1 has the 10 GbE ports (on GTH, lower power)

SHEET-2

SOM_B2B_2



BTB 2 has the high-speed TURFIO stuff

SHEET-3

SOM_B2B_3



BTB 3 is unused

SHEET-4

SOM_B2B_4



BTB 3 is unused

SHEET-5

Mechanical stackup works like:
TURF (5 mm spacer) SOM

TURFIO_1

CLKA_P
CLKA_N
RXCLKA_P
RXCLKA_N
TXCLKA_P
TXCLKA_N
COUTA_P
COUTA_N
CINTIOA_P
CINTIOA_N
CIN0A_P
CIN0A_N
CIN1A_P
CIN1A_N
CIN2A_P
CIN2A_N
CIN3A_P
CIN3A_N
CIN4A_P
CIN4A_N
CIN5A_P
CIN5A_N
CIN6A_P
CIN6A_N
SPAREA_P
SPAREA_N
(NULL)
TRXA
TCTRIA
TGPI01A
MGTRXA_P
MGTRXA_N
(NULL)
(NULL)
CALA_P
CALA_N

SHEET-6

TURFIO_2

CLKA_P
CLKA_N
RXCLKA_P
RXCLKA_N
TXCLKA_P
TXCLKA_N
COUTA_P
COUTA_N
CINTIOA_P
CINTIOA_N
CIN0A_P
CIN0A_N
CIN1A_P
CIN1A_N
CIN2A_P
CIN2A_N
CIN3A_P
CIN3A_N
CIN4A_P
CIN4A_N
CIN5A_P
CIN5A_N
CIN6A_P
CIN6A_N
SPAREA_P
SPAREA_N
(NULL)
TRXA
TCTRIA
TGPI01A
MGTRXA_P
MGTRXA_N
(NULL)
(NULL)
CALA_P
CALA_N

SHEET-7

TURFIO_3

CLKA_P
CLKA_N
RXCLKA_P
RXCLKA_N
TXCLKA_P
TXCLKA_N
COUTA_P
COUTA_N
CINTIOA_P
CINTIOA_N
CIN0A_P
CIN0A_N
CIN1A_P
CIN1A_N
CIN2A_P
CIN2A_N
CIN3A_P
CIN3A_N
CIN4A_P
CIN4A_N
CIN5A_P
CIN5A_N
CIN6A_P
CIN6A_N
SPAREA_P
SPAREA_N
(NULL)
TRXA
TCTRIA
TGPI01A
MGTRXA_P
MGTRXA_N
(NULL)
(NULL)
CALA_P
CALA_N

SHEET-8

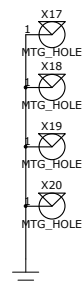
TURFIO_4

CLKA_P
CLKA_N
RXCLKA_P
RXCLKA_N
TXCLKA_P
TXCLKA_N
COUTA_P
COUTA_N
CINTIOA_P
CINTIOA_N
CIN0A_P
CIN0A_N
CIN1A_P
CIN1A_N
CIN2A_P
CIN2A_N
CIN3A_P
CIN3A_N
CIN4A_P
CIN4A_N
CIN5A_P
CIN5A_N
CIN6A_P
CIN6A_N
SPAREA_P
SPAREA_N
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TRXA
TCTRIA
TGPI01A
MGTRXA_P
MGTRXA_N
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(NULL)
CALA_P
CALA_N

SHEET-9

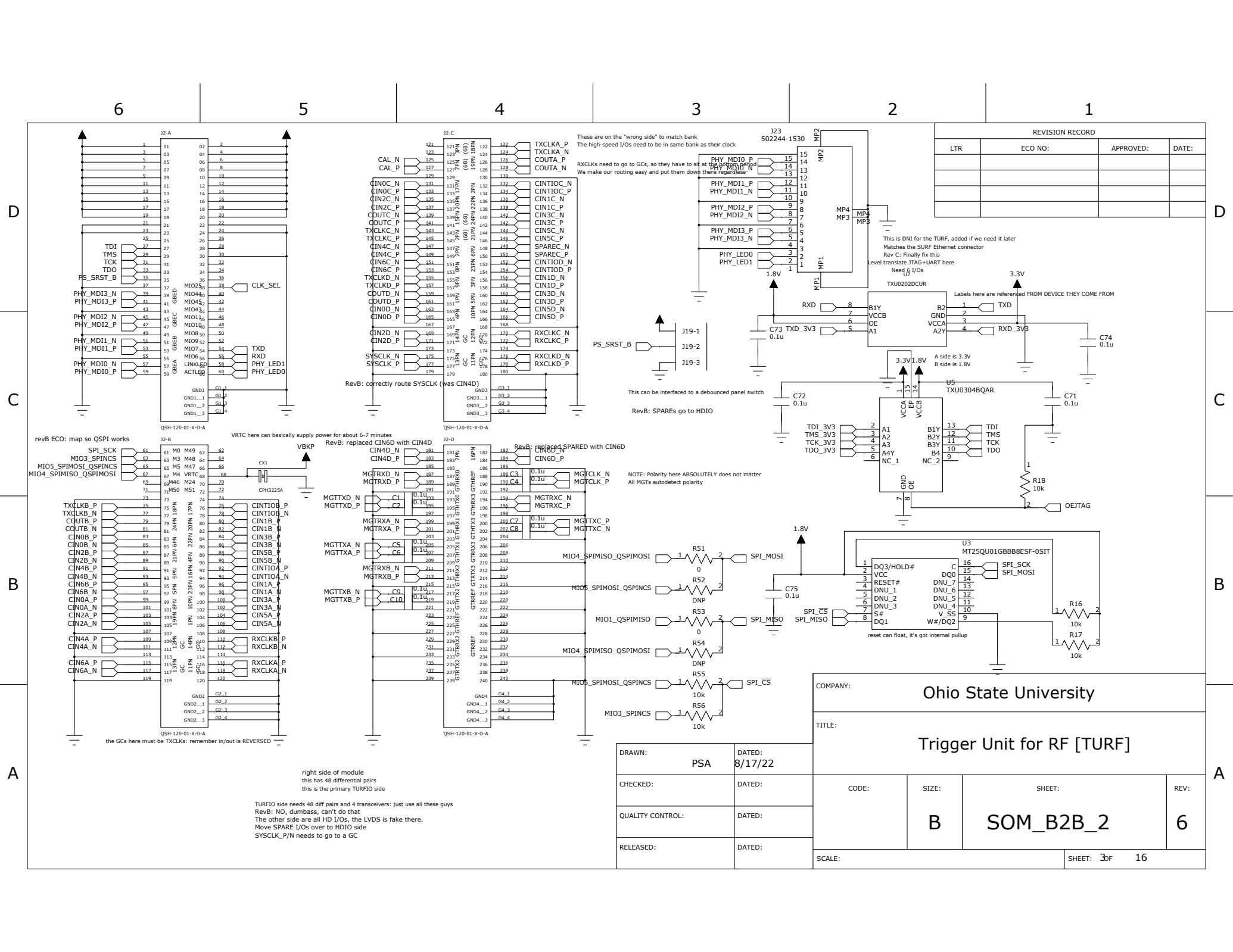
REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:



COMPANY:		Ohio State University	
TITLE:			
Trigger Unit for RF [TURF]			
CODE:	SIZE:	DRAWING NO:	REV:
	B	A	6
SCALE:		SHEET: 1 of 16	

DRAWN: PSA	DATED: 8/17/22
CHECKED:	DATED:
QUALITY CONTROL:	DATED:
RELEASED:	DATED:



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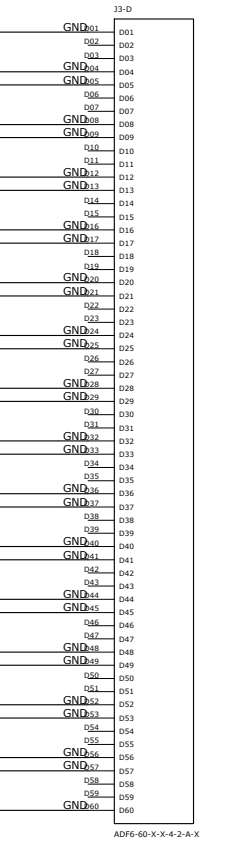
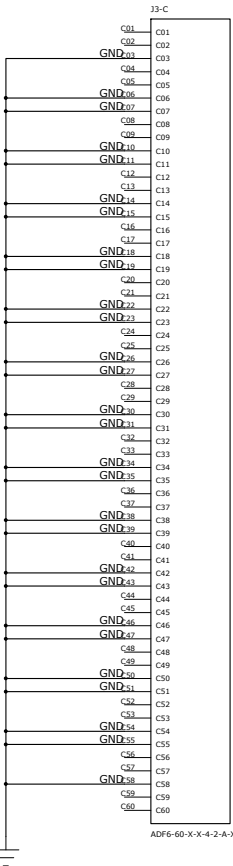
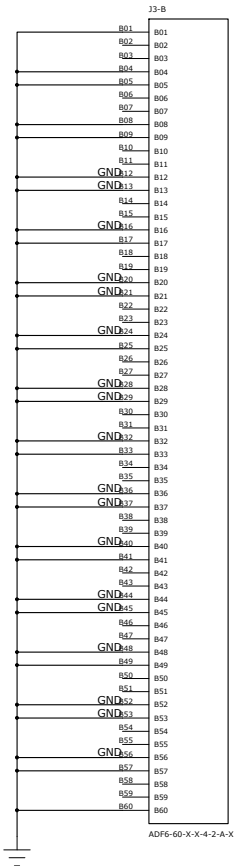
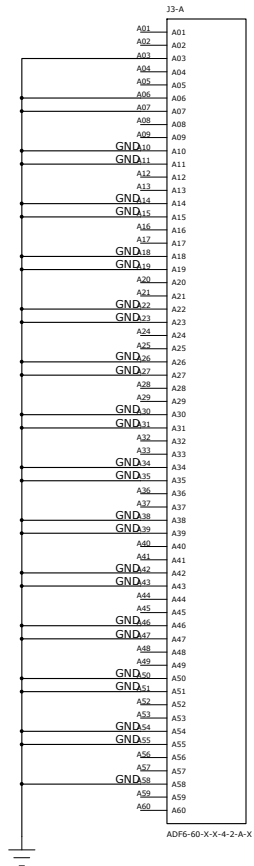
3

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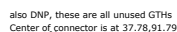
1

This will be DNP, it's all GTYs
Center of connector on SOM is at 37.79, 84.47

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY:				Ohio State University							
TITLE:				Trigger Unit for RF [TURF]							
DRAWN:		DATED:		CODE:		SIZE:		SHEET:		REV:	
PSA		8/17/22									
CHECKED:		DATED:									
QUALITY CONTROL:		DATED:									
RELEASED:		DATED:									
SCALE:				SHEET: 4 of 16							

REVISION RECORDLTRECO NO:

APPROVED:

DATE:DRAWN:PSA

DATED:
8/17/22

CHECKED:

DATED:

QUALITY CONTROL:

	DATED:
--	--------

RELEASED:

	DATED:
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COMPANY:

Ohio State University

TITLE:

Trigger Unit for RF [TURF]CODE:

SIZE:

SHEET:RFV:B

SOM B2B 4

6

SCALE:

SHEET: 50F	16
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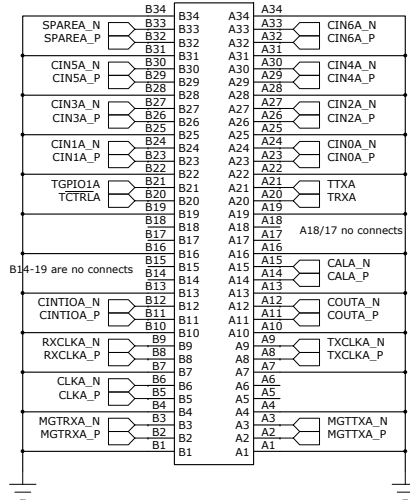
2

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REVISION RECORD			
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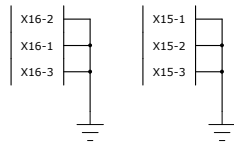
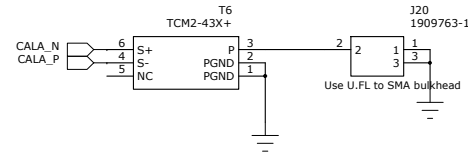
These have to follow the TURF SIDE SPEC

J5
75586-0002



TURF cable detect is via TCTRL
TURF side weak pulls up, TURFIO strong pulls down

N.B.: If we try to loop a cable back, we MUST tristate TGPI01, TCTRL, and TTX
And also don't hook up multiple CALs.



COMPANY:		Ohio State University	
TITLE:		Trigger Unit for RF [TURF]	
CODE:	SIZE:	SHEET:	REV:
	B	TURFIO_1	6
SCALE:		SHEET: 6 of	16

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LTR

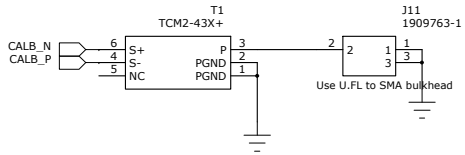
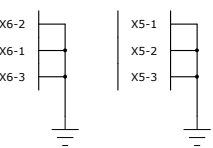
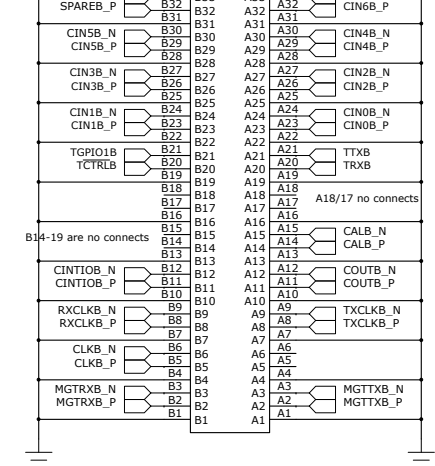
ECO NO:

APPROVED:

DATE:

J6

75586-0002



COMPANY: Ohio State University

TITLE: Trigger Unit for RF [TURF]

DRAWN: PSA	DATED: 8/17/22
CHECKED:	DATED:
QUALITY CONTROL:	DATED:
RELEASED:	DATED:

CODE:	SIZE: B	SHEET: TURFIO_2	REV: 6
SCALE:	SHEET: 7 of 16		

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REVISION RECORD

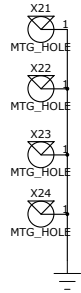
LTR	ECO NO:	APPROVED:	DATE:

TURFIO 3 and 4 connect via riser PCB
Each has 16 diff pairs + 5 GPIOs = 37 signals

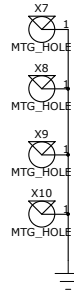
We need 32 pairs + 16 grounds to shield = 80 total there
Plus 8 more single ended + 4 grounds on either side = 92 total

This is 80 pin: we need
16 pairs + 5 single ended
So total is 32 pairs, 10 single ended = 74 total (6 open)

side panel mount



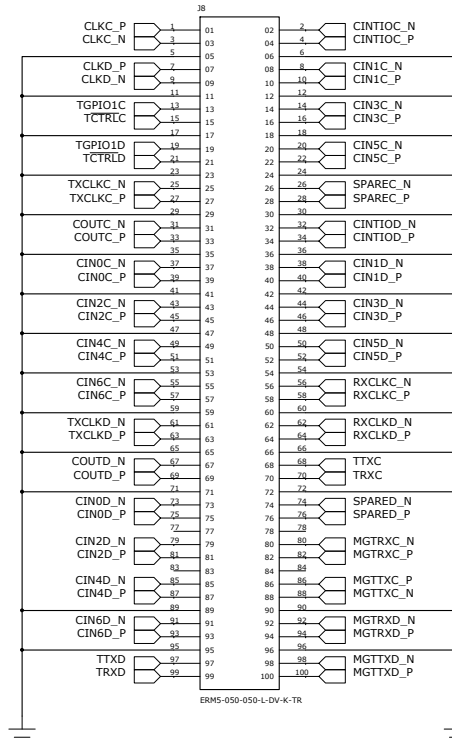
mounting holes for secondary board



These should go:
CLKC/D

RXCLK
IN
OUT0
OUT2
OUT4
OUT6

Clocks here end up requiring a 180 turn, but easier on the TURF than the TURFXB



COMPANY:

Ohio State University

TITLE:

Trigger Unit for RF [TURF]

DRAWN:

PSA

DATED:

8/17/22

CHECKED:

DATED:

QUALITY CONTROL:

DATED:

RELEASED:

DATED:

CODE:

SIZE:

SHEET:

REV:

B

TURFIO_3

6

SCALE:

SHEET: 8 of 16

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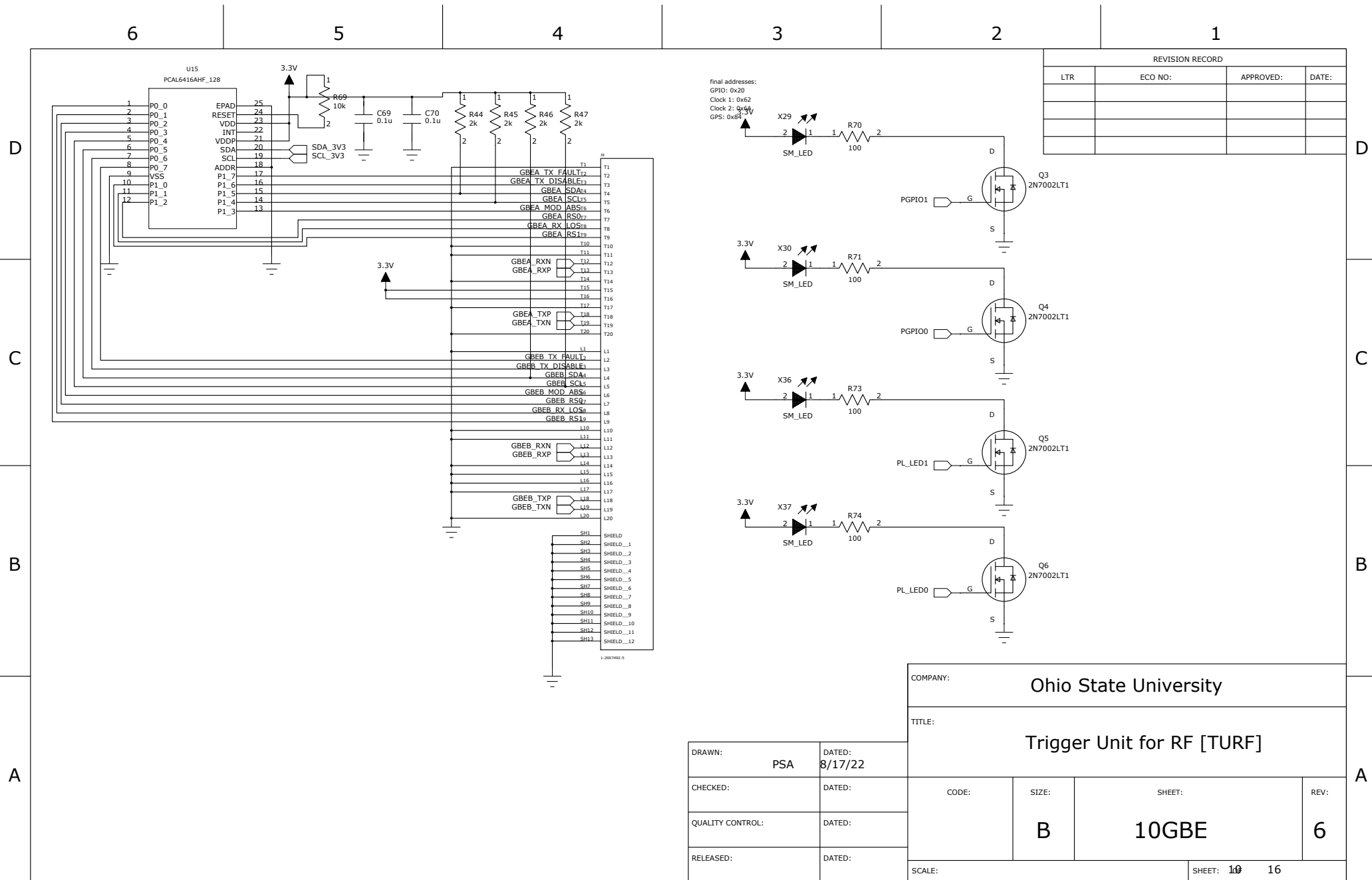
B

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There is no TURFIO_4; TURFIO 3 & 4 connectors are attached via daughterboard

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

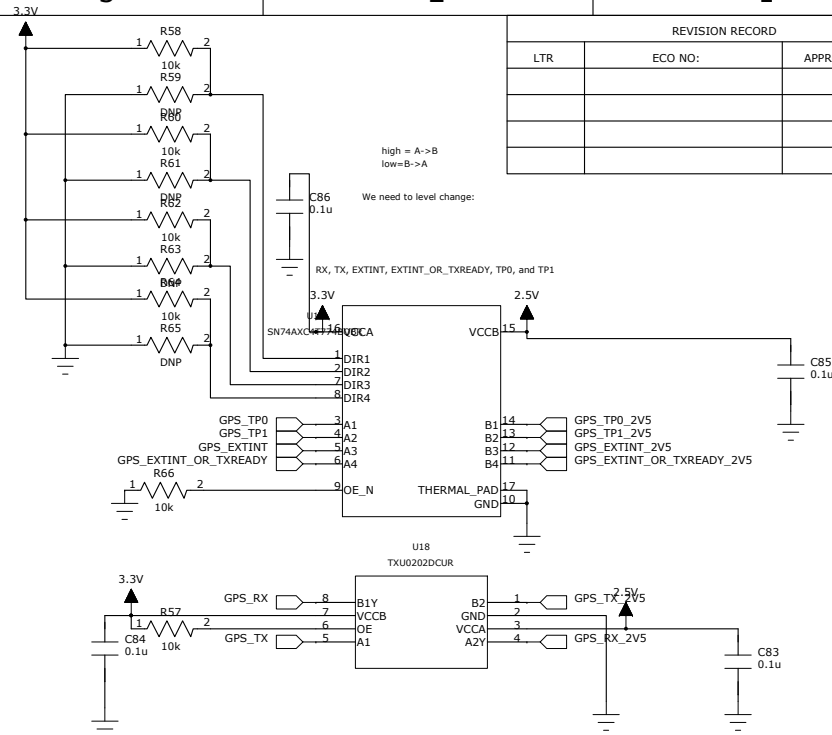
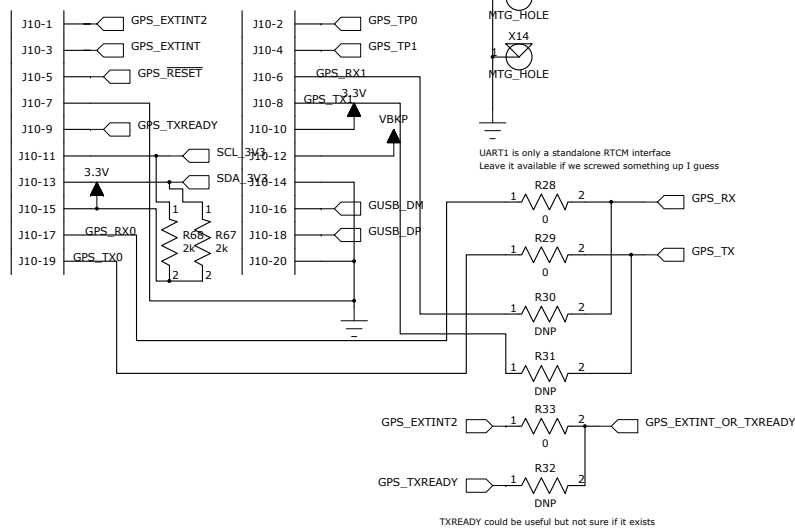
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CHECKED:		DATED:		CODE:	SIZE: B	SHEET: TURFIO_4		REV: 6
QUALITY CONTROL:		DATED:						
RELEASED:		DATED:						
				SCALE:			SHEET: 9 of 16	





For the GPS, we want:
 USB (5 pins total w/ground shield)
 6 interface inputs (just grab em all)
 reset
 vcc (2, pwr + ret)
 vbkp (2, pwr + ret)
 timepulse (x2)
 this is 18 total
 do a 2x10 interface?

Backup current is $36 \mu\text{A} + 2.5 \mu\text{A}$
The odds/evens flop here vs TINYZEDF9T because it faces down, we face up



REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

DRAWN: PSA		DATED: 8/17/22		Trigger Unit for RF [TURF]			
CHECKED:		DATED:		CODE:	SIZE: B	SHEET: GPS	REV: 6
QUALITY CONTROL:		DATED:					
RELEASED:		DATED:					
				SCALE:		SHEET: 1 of 16	

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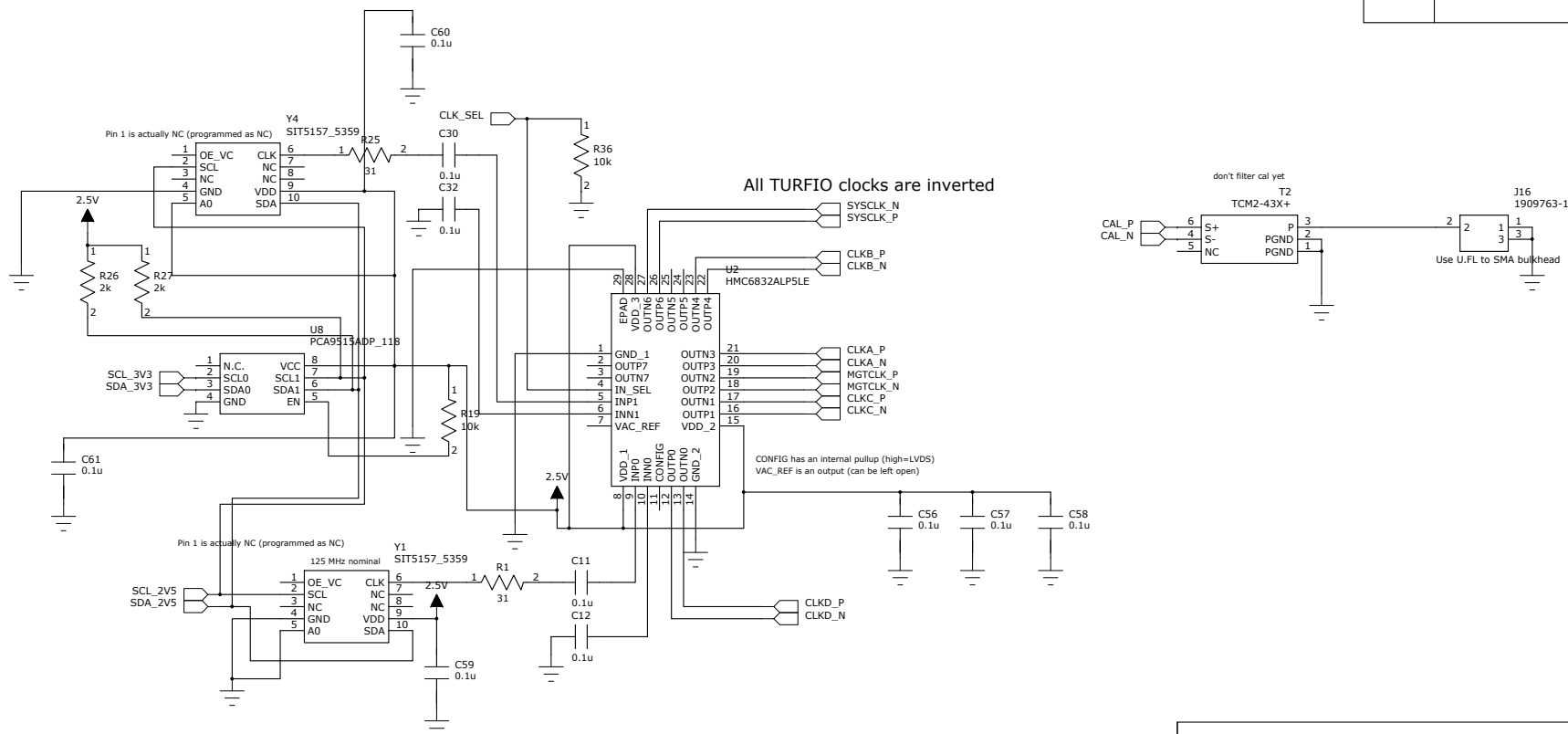
B

B

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REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY:		Ohio State University			
TITLE:		Trigger Unit for RF [TURF]			
DRAWN:	PSA	DATED:	8/17/22	CODE:	SIZE:
CHECKED:		DATED:			
QUALITY CONTROL:		DATED:			
RELEASED:		DATED:			
SCALE:				SHEET: 12	16

CLOCK

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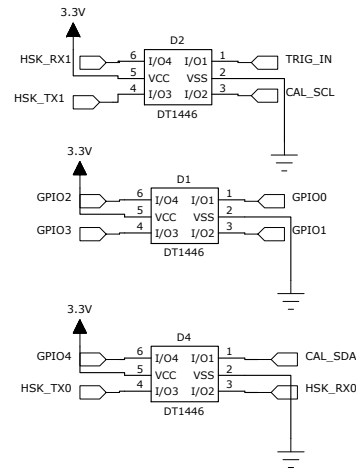
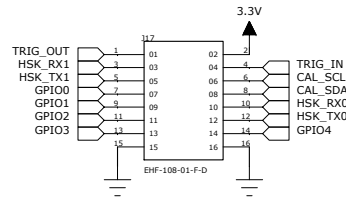
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REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:



COMPANY:

Ohio State University

TITLE:

Trigger Unit for RF [TURF]

DRAWN:

PSA

DATED:

8/17/22

CHECKED:

DATED:

QUALITY CONTROL:

DATED:

RELEASED:

DATED:

CODE:

SIZE:

SHEET:

REV:

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HSK

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SCALE:

SHEET: 13 16

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D

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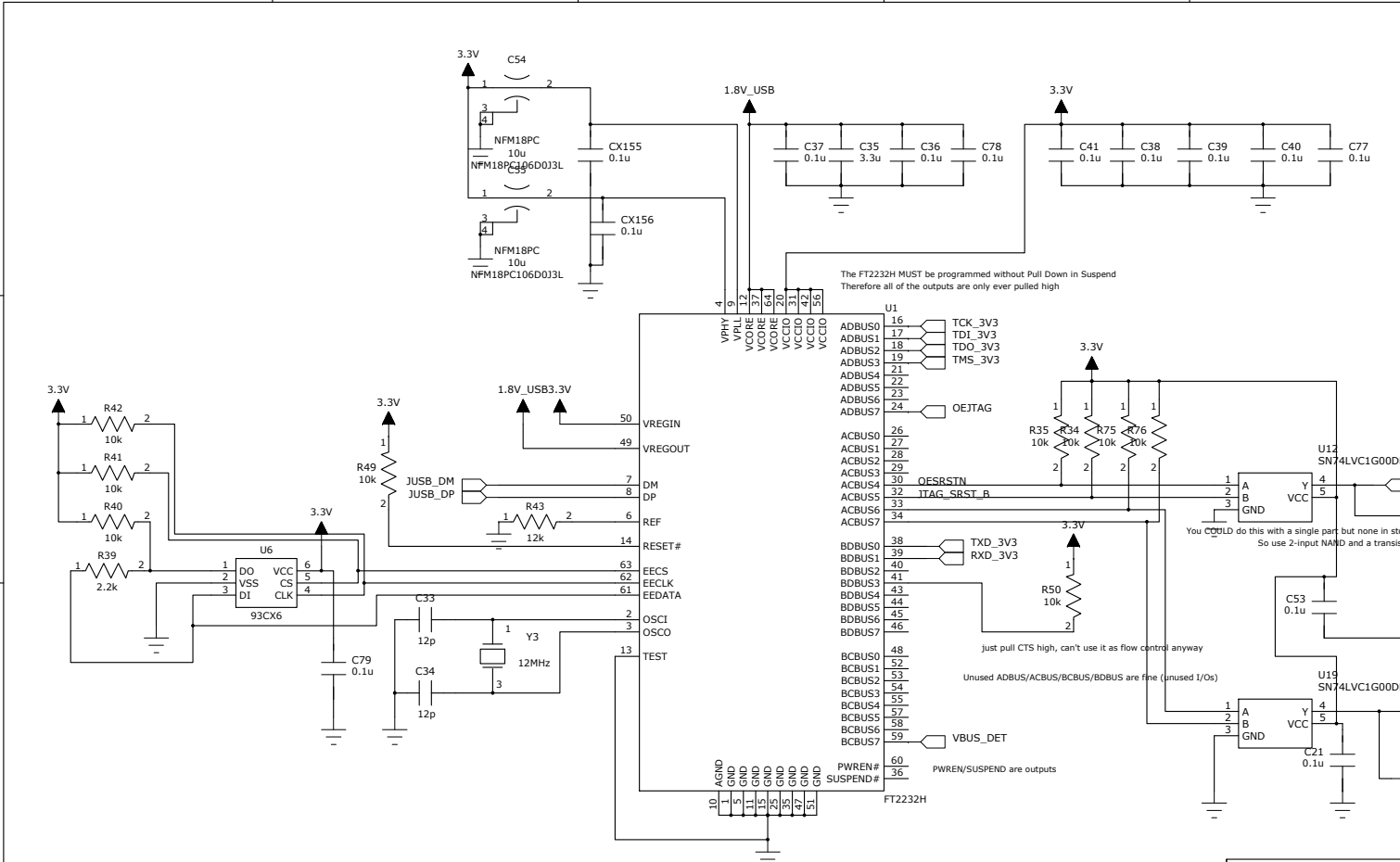
B

B

A

A

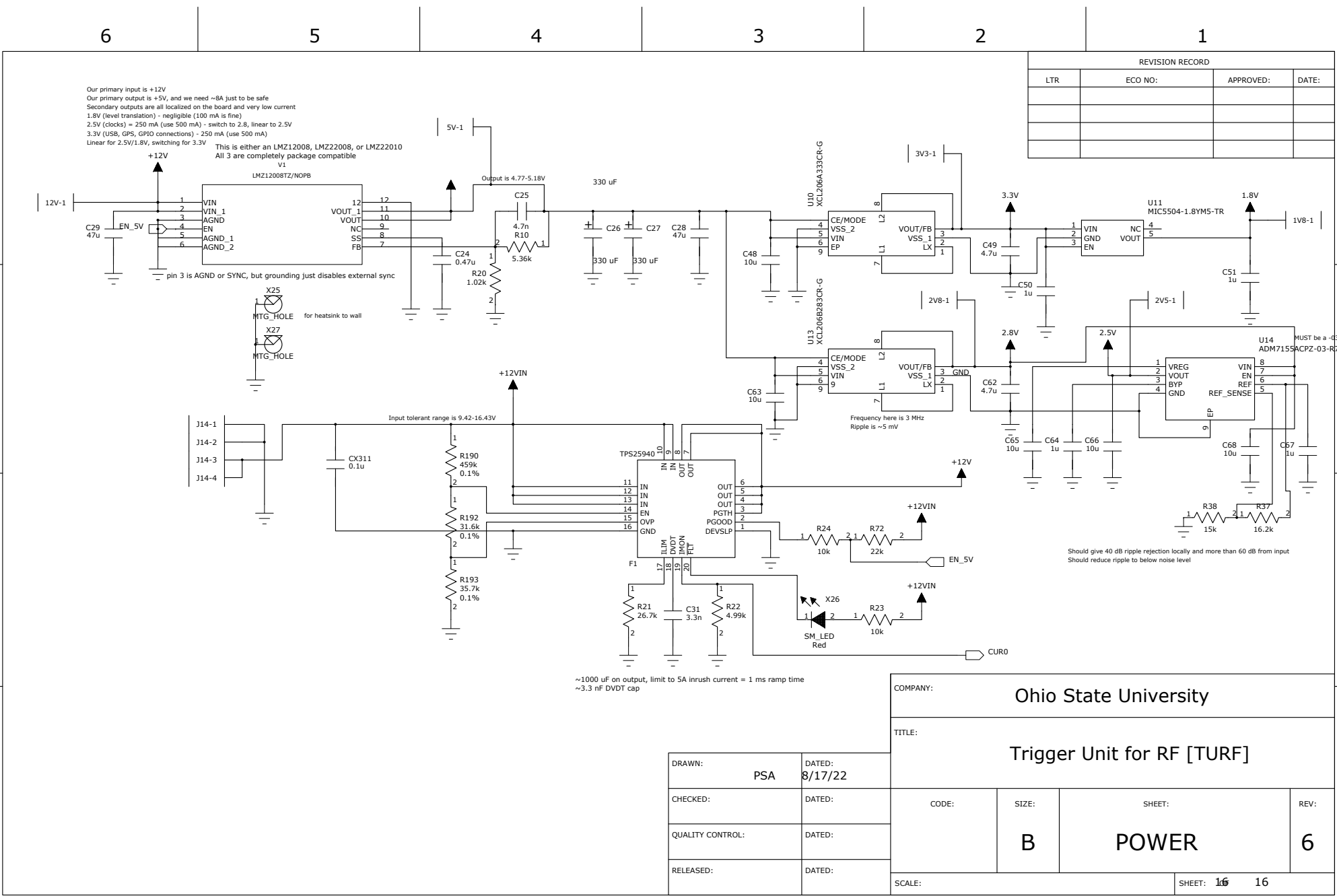
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: Ohio State University			
TITLE: Trigger Unit for RF [TURF]			
DRAWN: PSA	DATED: 8/17/22	CODE:	SIZE:
CHECKED:	DATED:	SHEET: B	
QUALITY CONTROL:	DATED:	JTAG	
RELEASED:	DATED:	6	
SCALE:		SHEET: 15 16	

D
C
B
A

D
C
B
A



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